

LC 1404: Number of steps to reduce a number in binary representation to one

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Original Handwritten Notes

 Handwritten notes for LC 1404

Method 1

Approach: Convert binary string to long long using `stoll`.

Then: `while (num != 1)`

- if odd $\rightarrow \text{num} = \text{num} + 1$
- if even $\rightarrow \text{num} = \text{num} >> 1$ (divide by 2)

Count no. of ops it took.

PROBLEM

A string can be very very large which might not fit even in `long long`.

Code: Method 1 (Using stoll)

This method works for smaller strings but fails on very large binary representations due to integer overflow limits.

```
class Solution {
public:
    int numSteps(string s) {

        long long num=stoll(s,0,2);
        int ans=0;

        while(num!=1){
            if(num&1){
                num=num+1;
            }

            else{
                num=num>>1;
            }

            ans++;
        }

        return ans;
    }
};
```

Method 2: Direct String Manipulation

HENCE

Need to operate directly on strings.

- for(int i = n-1; i > 0; i--) {
- if (s[i] == '0') {
// even, so divide by 2, already going left, so just count the ops ans++; }
- else {
// last bit 1, odd 1, results to 0 and gives a carry 1
// propagate that carry to left

Code: Method 2 (Optimal)

This approach bypasses the integer limits by manually manipulating the string and propagating the carry bit as needed.

```
class Solution {
public:
    int numSteps(string s) {

        int ans=0;
        int n=s.size();

        for(int i=n-1;i>0;i--){
            //dont go till first index
            //bcz if baad me 1 ho jaayega to +1 add krke 10 ho jaayega
            //lekin 1 pe rukna tha
            if(s[i]=='0'){
                ans++;
            }

            else{

                //odd -> +1 and then divide by 2
                ans+=2;
                int j=i;
                while(j>=0 && s[j]=='1'){
                    s[j]='0';
                    j--;
                }

                if(j>=0){
                    s[j]='1';
                }

                else{

                    //maan le koi 0 nhi tha
                    //11
                    //to 100 ban jaayega
                    //i.e a new 1

                    ans+=i;
                    break;
                }
            }
        }
    }
}
```